

Review



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The role of oxidative stress in postcopulatory selection

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Two decades ago, von Schantz *et al.* (von Schantz T, Bensch S, Grahm M, Hasselquist D, Wittzell H. 1999 Good genes, oxidative stress and condition-dependent sexual signals. *Proc. R. Soc. B* **266**, 1–12. (doi:10.1098/rspb.1999.0597)) united oxidative stress (OS) biology with sexual selection and life-history theory. This set the scene for analysis of how evolutionary trade-offs may be mediated by the increase in reactive molecules resulting from metabolic processes at reproduction. Despite 30 years of research on OS effects on infertility in humans, one research area that has been left behind in this integration of evolution and OS biology is postcopulatory sexual selection—this integration is long overdue. We review the basic mechanisms in OS biology, why mitochondria are the primary source of ROS and ATP production during oxidative metabolism, and why sperm, and its performance, is uniquely susceptible to OS. We also review how postcopulatory processes select for antioxidation in seminal fluids to counter OS and the implications of the net outcome of these processes on sperm damage, sperm storage, and female and oocyte manipulation of sperm metabolism and repair of DNA to enhance offspring fitness.

This article is part of the theme issue 'Fifty years of sperm competition'.

We believe that the role of reactive oxygen species in postcopulatory sexual selection deserves immediate empirical assessment.

Dowling and Simmons [7, p.1742]

We, the authors of the present article, agree with Dowling and Simmons.

1. Introduction: oxidative stress effects on sperm and their competition

Oxidative stress (OS) occurs when levels of reactive oxygen species (ROS), primarily generated by the electron transport chain (ETC), exceed those necessary for cellular signalling and overwhelms the detoxification capacity of a biological system [1]. Excessive ROS damages DNA, proteins and membrane lipids, ultimately disrupting normal cell function [1,2]. von Schantz *et al.* [3] was the first to recognize the integral role OS played in sexual selection theory. Despite the success of von Schantz *et al.*'s insight in precopulatory sexual selection, corresponding research into the direct connections between OS and postcopulatory processes has been slow to launch, despite copious evidence linking oxidative damage to infertility in men dating back to the late 1980s [4,5]. From an evolutionary perspective, this is unfortunate, since selection on haploid states is expected to be particularly strong given that there is nowhere for recessive alleles to hide. In light of this, Peters *et al.* [6] laid out the logical argument that sperm function might be compromised by OS. Dowling & Simmons [7] then explicitly pointed out that OS is likely to be

critically important in postcopulatory selection, since oxidative damage may reduce competitive fertilization success, even with minimal sperm damage [1,7].

Sperm competition (SC) occurs when ejaculates of different males compete for the fertilization of ova [8]. This process is fundamentally a game of numbers: all else being equal, he who gets the most sperm to sites of fertilization has the highest probability of fertilizing the ova [9]. However, in internal fertilizers, less than 0.01% of inseminated sperm make it to the fertilization sites [10], and because mating and ovulation are temporally separated, sperm must often remain viable within the oviduct for months to years before fertilization ([11], the latter in e.g. queens of social insects). Thus, to compete for fertilization, a male must have high-functioning sperm that can reach the ova more quickly and remain viable for longer than other males—processes that may select for sperm with an optimized longevity–speed trade-off [12]. Furthermore, if the spermatozoa of particular males have different degrees of molecular damage (e.g. DNA), then selection may more strongly favour polyandry as a bet-hedging mechanism to favour ‘good’ sperm. Sperm damage may select for female processes that filter sperm that may be inferior at fertilization, or poor at building a viable zygote. Such sperm damage would then, potentially, also select for female or oocyte processes that repair malfunctioning sperm.

In this review, we first describe the sources of oxidative damage and mitochondrial function. Next, we address the evidence for OS affecting sperm function in covering some of the relatively sparse behavioural ecology literature on the subject. Finally, we explore the selection of females (or eggs) to repair damaged sperm DNA, heavily researched in human assisted reproductive technologies (ART), such as intracytoplasmic sperm injections (ICSI), but largely overlooked in most other taxa. We end with a forward-looking section that highlights areas that are ripe for research but have yet to be explored, and thus cannot be reviewed.

2. Reactive oxygen species and mitochondrial function

In humans, 68% of the variation in sperm motility is explained by variation in mitochondrial ROS production [13]. Although there are other sources of ROS (e.g. generated through the pentose phosphate pathway, metal-catalysed oxidation and the oxidative bursts of leucocytes [4]), here we refer to all ROS and reactive molecules collectively as ‘ROS’.

In most species [14], mitochondria function to power sperm movement through the production of ATP but are also the primary sources of ROS. Electrons passing along protein complexes in the ETC are coupled with the movement of protons across the inner mitochondrial membrane, creating an electrochemical gradient. This gradient drives protons back into the mitochondrial matrix through ATP synthase, generating ATP through oxidative phosphorylation (OXPHOS) [15]. As electrons are transferred to and through the ETC, some leak and react with molecular oxygen to produce superoxide and other ROS molecules in downstream reactions (e.g. superoxide O_2^- ; hydrogen peroxide H_2O_2 ; hydroxyl radical $\cdot OH$) [15]. Steeper electrochemical gradients (i.e. higher proton motive force) established across the inner mitochondrial membrane results in greater ROS production [15,16] because

respiration slows and electrons accumulate along the ETC instead of being passed to oxygen, which combines with two protons to form water [15]. The expected interrelations between ATP and ROS production are not completely understood [16], but dissipation of the proton motive force, as a result of high ATP synthesis, would result in an inverse relationship between ATP and ROS production because electrons can move freely through the ETC to be accepted by oxygen, producing ATP. On the other hand, a positive relationship between ROS and ATP might also be expected; uncoupling proteins and passive proton leak can change the electrochemical potential, resulting in inefficient ATP production. The relationship between ATP and ROS is further complicated by the fact that they also depend on sperm energy balance and activity and substrate availability (e.g. ADP [15,16]). Balancing these factors is expected to play a pivotal role during SC because ‘speed’ may come at a cost to longevity [17]. Genetic and environmental variation should mediate these trade-offs given that the efficiency of OXPHOS in mitochondria, and therefore ROS and ATP production, is known to depend on both cytonuclear genetic variation and the specific environment that mitochondria experience [13,18–20].

Non-neutral mitochondrial (mt) DNA variants that affect mitochondrial function are likely to have strong effects on sperm performance given that mtDNA is inherited (almost exclusively) maternally, and, thus, exposed to selection only through females [20,21]. For example, mitochondrial genetic variation in genes encoding tRNA^{Arg} in chickens, *Gallus domesticus*, are known to increase the presence of aberrant mitochondria in sperm, resulting in reduced sperm oxygen consumption and a 50% reduction in sperm motility [19]. Separate work using two different chicken strains has shown that higher sperm motility is associated with both higher levels of ATP and lower ROS as a result of varying OXPHOS activity [22]. Similarly, mtDNA polymorphisms across mice containing homogeneous nuclear DNA backgrounds have also been linked to reduced sperm performance, but the mechanisms remain unclear given that ATP production did not differ across strains [23]. Such deleterious mutations that affect sperm can accumulate in a population if they are neutral to females, resulting in reduced male fertility and fitness (i.e. the ‘mother’s curse’) [20,21]. However, strong selection for compensatory mutations in nuclear genes can counter these adverse effects in males [21]. Understanding how gene–environment interactions affect mitochondrial and sperm function may be essential in explaining hybrid infertility across genetic lineages resulting in cytonuclear genetic incompatibility/mismatch [24]. In subtler cases, within a population or species, the cytonuclear genetic variation that impacts OXPHOS function is also predicted to affect sperm performance and thus among-male differential competitive fertilization success overall, but studies of these links are still rare. How environments shift sperm bioenergetics and competitiveness across different genotypes, mitochondrial haplotypes, female internal- and external ambient environments is still an understudied research area [12].

3. Sperm and oxidative damage

Spermatozoa are terminally differentiated, generally lack significant cytoplasm and have limited antioxidative defences; thus, spermatozoa are mostly unable to limit or

repair molecular damage to DNA, proteins and membrane lipids subsequent to spermiogenesis [25,26]. In addition, sperm membranes are composed of polyunsaturated fatty acids (PUFAs) with a high proportion of easily oxidized double-bonds, which makes them particularly vulnerable to oxidative damage [27]. Once membranes are oxidized, they rapidly generate a damaging cascade of radical chain reactions, further disrupting cell membranes [28]. Oxidative damage degrades overall sperm function and reduces the proportion of progressively motile sperm, parameters strongly associated with male infertility in humans [5,13,26]. Such factors are expected to generate strong selection on male reproductive traits, with variation across taxa (and mating systems within taxa) in how to mitigate sperm damage and increase reproductive success [7].

DNA damage is particularly common under oxidative stress and has been observed in testicular, epididymal and ejaculated sperm, with most information available for humans and mammal models [25]. Based on data for these taxa, DNA fragmentation is characterized by both single (SSB) and double DNA strand breaks (DSB) and may arise for several reasons, including oxidative attacks. DSBs are extremely malignant lesions that can lead to genomic instability and cell death if not properly repaired, and again, sperm are not capable of such repair. These findings, in addition to strong positive correlations between DNA damage and 8-OH-dG expression (a modified nucleoside base, excreted upon DNA repair [29]), suggest that sperm DNA damage is frequently a product of OS. DNA damage and repair are age-dependent in adult tissues [30] and immature sperm produce much higher levels of ROS [25]. This ROS production in young sperm can induce greater DNA damage in mature sperm, effects produced after spermiation from the seminiferous tubules, and after ejaculation [25]. Some studies have shown increased fragmentation in men over 35, while others have not, and similar contradictory results come from work on copulatory abstinence, with some confirming effects, others not (reviewed in [25]). If widespread, OS effects resulting from copulatory abstinence could affect selection arising from SC and cryptic female choice [25], for instance, if copulation frequency affects fertilization success.

4. Not all is lost if there is ROS

ROS also have positive effects on cell signalling and sperm function. The capacitation of sperm required to penetrate and fertilize an egg in mammals is triggered by ROS [31]. However, even when produced at damaging levels, ROS can be quenched by endogenous antioxidant enzymes, such as superoxide dismutase (SOD), catalase (CAT) and glutathione peroxidase (GPx) [2,32]. Expression of these endogenous antioxidants and mitochondrial replication (which can lower ROS) can be further stimulated by ROS [33]. Of these endogenous antioxidants, SOD and CAT require metal cofactors (e.g. Zn, Cu, Fe), or in the case of GPx, selenium [34], thus making these components susceptible to dietary deficiencies for antioxidation and not strictly endogenously regulated [35]. The same can be argued for potential antioxidation by carotenoids, which is often an abundant pigment in sexually selected ornaments [36]. Non-enzymatic antioxidants, e.g. glutathione, uric acid, ascorbic acid, tocopherols [1], provide links to condition-dependence of sperm function, but these are mostly untested. Individuals

can differ in their capacity to resist and repair damage from OS [2], with ROS and endogenous antioxidant levels showing moderate to high heritability ($0.45 < h^2 < 0.59$, across taxa [37–39]). This ability to resist damage may translate to the germline and the seminal fluid, which contains and supports sperm, but few studies have tested this assumption. Despite this lack of knowledge, we do have evidence that variation in sperm performance is heritable (e.g. [40,41]), and sperm and ejaculate quality has been shown to evolve in response to selection [42], suggesting ample capacity for evolutionary responses.

5. Sperm performance, ROS and seminal antioxidation

What constitutes high sperm ‘performance’ or ‘quality’ is dependent on species and mating context, but sperm velocity, motility, viability and longevity are predicted to be linked to male competitive reproductive success [43]. The production of more sperm requires increased cell divisions and metabolic activity in the testes [44], which may be accompanied by an increase in ROS production [45,46]. Even in relatively monogamous species with short-term sperm storage, such as humans [47], oxidative damage of sperm strongly influences fertilization and male reproductive success [48].

Although sperm have some level of antioxidant capacity (i.e. peroxiredoxin—[49]), the ejaculate fluid is the primary source of protection from oxidative damage and essential for fertilization success by facilitating normal sperm function [50].

There is an extensive literature on antioxidant treatment effects on infertility in men with mixed results (reviewed in [51]). These conclude positive effects of antioxidant treatments on, for example, sperm motility and that treatments may elevate live birth rate (but data quality for live births was considered ‘low’, owing to complex interpretation of many predictors (our interpretation)). In addition, ROS aside, low-mobility sperm in domestic turkeys, *Meleagris gallopavo*, have a higher rate of early embryonic death, but genetic and mechanistic underpinnings to this finding are unknown [52]. Similar effects on sperm swimming performance were confirmed to be genomic by Alavioon *et al.* [53], but judgement is suspended regarding which specific genes predicted motility effects. Remarkably, these researchers also show that selection for long-lived sperm within ejaculates reduces reproductive ageing in the resulting offspring, a process that is often associated with ROS exposure through a number of pathways (including telomere attrition; e.g. [54]). ROS were never measured in this work, but ROS and OS have long been known to be involved in regulating gene expression [55]. This calls for analyses of interactive effects linking ROS and OS to carry-over effects on offspring fitness—such as life history modifying effects—by ROS across haploid and diploid phases, and to what degree such processes are initiated, and targets of selection, as early as gamete interaction and fertilization.

Age-related declines in endogenous antioxidant capacity of red junglefowl, *Gallus gallus*, have been associated with an increase in oxidative damage and a decline in sperm quality [56]. Recent work demonstrates that sperm age negatively affects sperm performance and the probability of paternity in SC between males [57] and even in competition between sperm within a single ejaculate [53]. Genetic and epigenetic

variation in sperm function in these experiments were even transgenerationally expressed in the sons produced in the zebra fishes (*Danio rerio*) used for these experiments. To what extent these age effects are ROS-dependent is not yet known.

In response to selection from ROS damage, testicular tissues may have more antioxidant defence than other tissues [58,59] and are ubiquitous components of seminal fluid across taxa: insects [60], birds [56], fish [61] and mammals [50]. For example, the testes of the Pacific cricket, *Teleogryllus oceanicus*, a species in which sperm viability, rather than the number sperm, most strongly influences paternity [62], has 3× the ROS levels and 1.4× the antioxidant capacity of muscle tissues used for courtship displays; nevertheless, the testes still had 2.5× more protein damage than muscle [46]. The effect of seminal antioxidation is clearly demonstrated when human sperm cells are washed free of seminal plasma—without their antioxidant cocktail, they rapidly undergo ROS-linked DNA damage (reviewed in [4]). The flip side experiment would be to expose the paternally contributed genes to antioxidants known to specifically target ROS within the mitochondria and this results in blockage of de-novo genetic damage in human spermatozoa [63]. In hamsters, *Mesocricetus auratus*, ablation of the accessory sex glands that produce a portion of the seminal fluid results in sperm oxidative DNA damage [64] that translates into lower fertilization success and lower offspring fitness.

Few studies combine measurements of oxidative status and competitive fertilization success. Three-spined sticklebacks (*Gasterosteus aculeatus*) fed carotenoid-supplemented diets have increased fertilization success [65] but this study did not test if ejaculate traits were affected and they were not in a competitive context. Knock-out mice, however, which are genetically deficient in SOD, have more oxidative damage and sperm with lower motility, velocity and competitive fertilization success than wild-type males [66], although evolutionary inferences from such data are hard to make. In painted dragon lizards, *Ctenophorus pictus*, superoxide measured in blood cells was negatively correlated with the proportion of motile sperm and ‘sperm performance’ (using a principle component analysis (PCA) of computer aided sperm analysis (CASA) parameters [67]) and positively correlated with body condition (more details in [67]). Given that superoxide and SOD were measured in blood, oxidative barriers between soma and germline may not be complete, or they may be species-specific [68]. Thus, OS at the organismal level may provide important insights into the condition-dependence of postcopulatory sexual selection [7]. For example, male Pacific crickets fed a diet supplemented with two antioxidant compounds, beta-carotene and α -tocopherol, sired significantly more offspring than controls after accounting for mating-order effects [69]. However, without measures of antioxidants, sperm traits or oxidative damage, it is difficult to interpret a direct causal role of ROS on paternity success.

Antioxidant treatment can also have null or negative effects on life-history traits [70], sperm performance [71,72] and fertility (reviewed in [73]). In humans, men on diets supplemented with beta carotene were believed more attractive by female subjects, but there was no effect on semen quality (based on WHO parameters [74]). Long-term supplementation with beta-carotene did not affect sperm traits (concentration, motility or velocity) at any dose used in the booroolong frog, *Litoria booroolongensis* [75]. The

interpretation of these results is challenging, given many alternative explanations, e.g. failure to assimilate the additional antioxidants, compensatory downregulation of production of endogenous antioxidants, or these dietary antioxidants not being limited in the baseline diet. The effect of dietary antioxidants can also interact with other dietary components, such as PUFAs, which can have strong positive effects on sperm function on their own [76]. For example, in guppies, *Poecilia reticulata*, a dietary reduction of omega 3 decreased sperm performance and the probability of paternity [77], whereas supplementing with a mixture of carotenoids interacted with PUFAs to enhance sperm viability [77,78]. Thus, to make evolutionary inference realistic, it is important to match the number and proportions of supplemented antioxidants with realistic values in the wild.

Several studies have manipulated social status or mating context, and thus possibly physiological stress, to test whether markers of OS correlate with sperm performance. In the bat *Carollia perspicillata*, sperm motility was unrelated to several antioxidant markers in the blood and whole ejaculate. OS also did not respond to experimental manipulation of the social context for SC [79]. However, sperm velocity was negatively correlated with levels of malondialdehyde (MDA: a product of oxidative lipid damage) in the ejaculate [79], showing ejaculate OS effects, whereas other studies did not [80,81]. Sperm swimming performance (using CASA) in the house sparrow, *Passer domesticus*, showed no effects of oxidants or antioxidants in semen (analysis by C.R.F. using data from [82]). In zebra finches, *Taeniopygia guttata*, treatment with a prooxidant (Diquat) decreased sperm velocity only in birds with the most colourful beaks, suggesting a trade-off between sexual ornamentation and sperm protection [83], and at dietary supplementation with carotenoids, more sperm were morphologically normal. These studies remind us that selection resulting from pre- and postcopulatory processes, and their condition-dependent expression, can result in trade-offs that affect sperm function, fitness and increased resource allocation towards antioxidation [7,84]. Manufacturing and maintaining high numbers of high-quality sperm, and the seminal fluid that sustains them, is energetically costly [85]. This selects for evolutionary trade-offs between pre- and postcopulatory selection [86]—and condition- and age-dependent expression of ejaculate traits [56,87,88]—that all might be affected and mediated by balancing oxidative status between the germline and soma.

6. ROS and OS in partners and parents: DNA damage and repair

Sperm form and function were long believed to be traits solely dictated at the diploid level (i.e. the soma or before the haploid state in spermatogenesis), and the realization that this is not the case put things into different perspectives [53,89–93]. Recent work demonstrates greater complexity in haploid biology than has previously been appreciated, such as covariation between sperm gene expression, motility and capacitation [94], and cryptic egg choice that runs counter to the diploid female’s preferences [95]. ROS insult likely has profound effects on many aspects of these processes of gamete biology, such as methylation and other epigenetic modification [96], potentially modifying gene expression. In the two-spotted cricket (*Gryllus bimaculatus*), sperm

stored in female storage organs have a 37% reduction in cell metabolic rate and a 42% reduction in ROS compared to sperm from fresh ejaculates—presumably facilitating sperm longevity, which may improve sperm competitive ability [59]. Reinhardt & Ribou [97] remind us that there are two routes to reduce oxidative sperm damage: interference with cell metabolism to reduce the formation of oxygen radicals, or the production of antioxidants. In bed bugs, *Cimex*, females alter the metabolism of sperm in their storage; when sperm metabolic rate is allowed to increase to the levels seen in males, females lay infertile eggs. This finding seems to agree with observations on *Apis mellifera* in which levels of proteins related to energy metabolism decline in stored compared to freshly ejaculated sperm [98]. *Apis* sperm can also switch between aerobic and anaerobic metabolism; aerobic, when they compete for access to the spermatheca and anaerobic, with the glycolytic metabolite GA3P serving as a key substrate for sperm survival during the several decades in storage following the queen's single mating (see [99] for details in social insects).

The alternative route to sperm protection, antioxidation, is taken in *A. mellifera*, by upregulating genes for anti-oxidation, apparently in parallel with the adaptation for sperm storage described above [100]; transcriptomic support for this, and upregulation of immune genes, have also been described in ant queens [101,102]. Similarly, upregulated oxidation occurs in *Anopheles gambiae* mosquitoes, in which a female mates once in her life and stores sperm for a lifetime (approx. 30 days) [103]. Females react to a male steroid component (20E) produced by male accessory glands that triggers heme peroxidase (HPX) production in females, which increases long-term sperm survival over multiple gonotrophic cycles. This seems to suggest no sexual conflict, since females elevate antioxidation in response to a male chemical compound, with the mutual interest of sperm survival. Similar agreement by selection—rather than conflict—is manifested when sperm DNA is ROS-damaged and then repaired by the oocyte during and after fertilization in humans that store sperm for short periods (1–7 days). Kamkar *et al.* [104], for example, show that in couples with repeated miscarriages, compared to fully fertile controls, semen had higher ROS levels, lower antioxidant capacity, higher sperm DNA fragmentation, lower sperm motility and higher miscarriage rate—the whole sequence from high ROS to low reproductive success. Observations from ART are probably the closest that we can come to the experimentation of these parameters; disregarding ROS, ICSI performed with sperm differing in DNA fragmentation show strong adverse effects of fragmentation on a combined index of embryo quality and the probability of pregnancy [105]. ROS–miscarriage relationships are, however, not straightforward. Many ROS-related factors affect female reproduction without necessarily being linked to ROS effects on sperm (e.g. reviewed in [106]). Furthermore, leucocytes in the ejaculate, bacterial infection and inflammation of the male reproductive system are significant contributors to male infertility [31], and the role of oxidative stress and damage owing to these factors is an exciting emergent field [107,108].

7. Future directions

Much is to be learned about how ROS and OS impact post-copulatory sexual selection. The dynamic relationship

between the rate of ROS production and the counter-balance of antioxidant protection means that researchers cannot rely on measurements of antioxidants or markers of damage alone to determine the oxidative status of an individual; ROS, or metabolites of ROS, and molecular damage need to be measured to ensure accurate interpretation of oxidative status (reviewed in [2,109]). Possible ways to test the causal relationship between OS and organismal fitness are manipulation with prooxidants to pose an oxidative challenge [110], provide dietary antioxidants to aid in quenching free radicals [111,112], and manipulation of endogenous oxidants [2,113]. Researchers employing these methods have myriad choices of pro- and antioxidants and must be cognizant of the possibility of overdosing animals. Thus, pilot studies are necessary to determine doses and potential side effects that may confuse interpretations (reviewed in [2,70,110–115]). Our limited knowledge about OS-induced damage and its fitness consequences during postcopulatory sexual selection still makes correlative studies worthwhile for uncovering natural variation in oxidative status that covaries with phenotypic traits. While there is much to be done yet, we describe particularly promising avenues for future research:

First, determining how much of the selection for sperm antioxidation arises from SC has never been examined, but is critical to understand the effect of OS on competitive fertilization success. Three testable predictions emerge from reviewing this research area: (i) sperm loss owing to storage should select for antioxidant production, and therefore may be particularly relevant in taxa with mating order effects and temporal separation of copulations and ovulations; (ii) antioxidant production should be particularly high in taxa with dissociated reproductive seasons, where sperm production and mating are decoupled (e.g. some squamate reptiles; [11]); and (iii) if the intensity of SC drives up sperm production, at some cost in terms of metabolic ROS production, then particularly high antioxidant levels are predicted in taxa with extreme gonadal somatic index (e.g. fairy-wrens, *Malurus splendens*; [116]). Testing these predictions can be done using comparative approaches both within and across species.

Second, the role of OS in mediating sexual conflict, and its relevance to sperm storage, is a particularly fascinating area in need of more work. We have previously identified higher levels of SOD in male compared to female dragon lizards and suggested SOD may not only be under sexual selection in males to rescue sperm from ROS but to force females to store sperm for longer than is optimal for females (sexual conflict; [117]). Under this scenario, we might predict that the degree of sperm storage in a population or species should be correlated with the magnitude of SOD differences between the sexes. However, if the repair of sperm DNA in females is widespread, non-discriminatory among individual males, and females primarily seem to be under selection to prolong sperm life, this may be hard to reconcile with sexual conflict (since female processes then act only to increase reproductive success in both sexes).

Lastly, ROS appear to be agents of mutation that produce high—sometimes extremely high—levels of fragmented DNA in spermatozoa. Assuming that these forces are non-directional with respect to coded and non-coded base pairs, at least some of these mutations will affect functional genes and potentially impact ideas like 'the sexy sperm' (heritability for winning SC) versus 'good sperm' (heritability for both

winning SC and producing fitter offspring; [118]). If so, ROS effects would be expected to be stronger for 'sexy sperm' since they would not even make it to fertilization. By contrast, beneficial effects on offspring from 'good sperm' scenarios would be suspended until the subsequent generation. ROS and OS effects on sperm gene expression represent a blank page in SC research. However, this area deserves greater attention, given the common gene-expression effects resulting from ROS and OS and the genomic and sperm morphometric effects on sperm motility.

8. Conclusion

In conclusion, we argue that ROS and OS need to be brought into the spotlight to address topical questions relevant to postcopulatory sexual selection theory—particularly given the radical effects ROS have on sperm competitive traits and longevity. Advances in technology for measuring mitochondrial function *in vitro* [119,120] and exciting new ways to manipulate oxidative status along with sophisticated tools for measuring sperm traits are providing fascinating opportunities for investigating mechanisms underlying OS-

mediated selection on SC, sexual conflict and cryptic female choice. At present, there are few studies of OS effects on sperm function/performance and competitive fertilization success in wild populations; even correlative studies have something to offer, by simply describing patterns to form baseline, foundational knowledge for future experimental studies.

Ethics. No animals were used.

Data accessibility. This article has no additional data.

Authors' contributions. All authors contributed to reviewing the literature, generation, and discussion of ideas and writing for this paper.

Competing interest. We declare we have no competing interests.

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